

Hawk-Hornet Invitational

November 15, 2025

Water Quality



School _____ Team # _____

Student Names _____

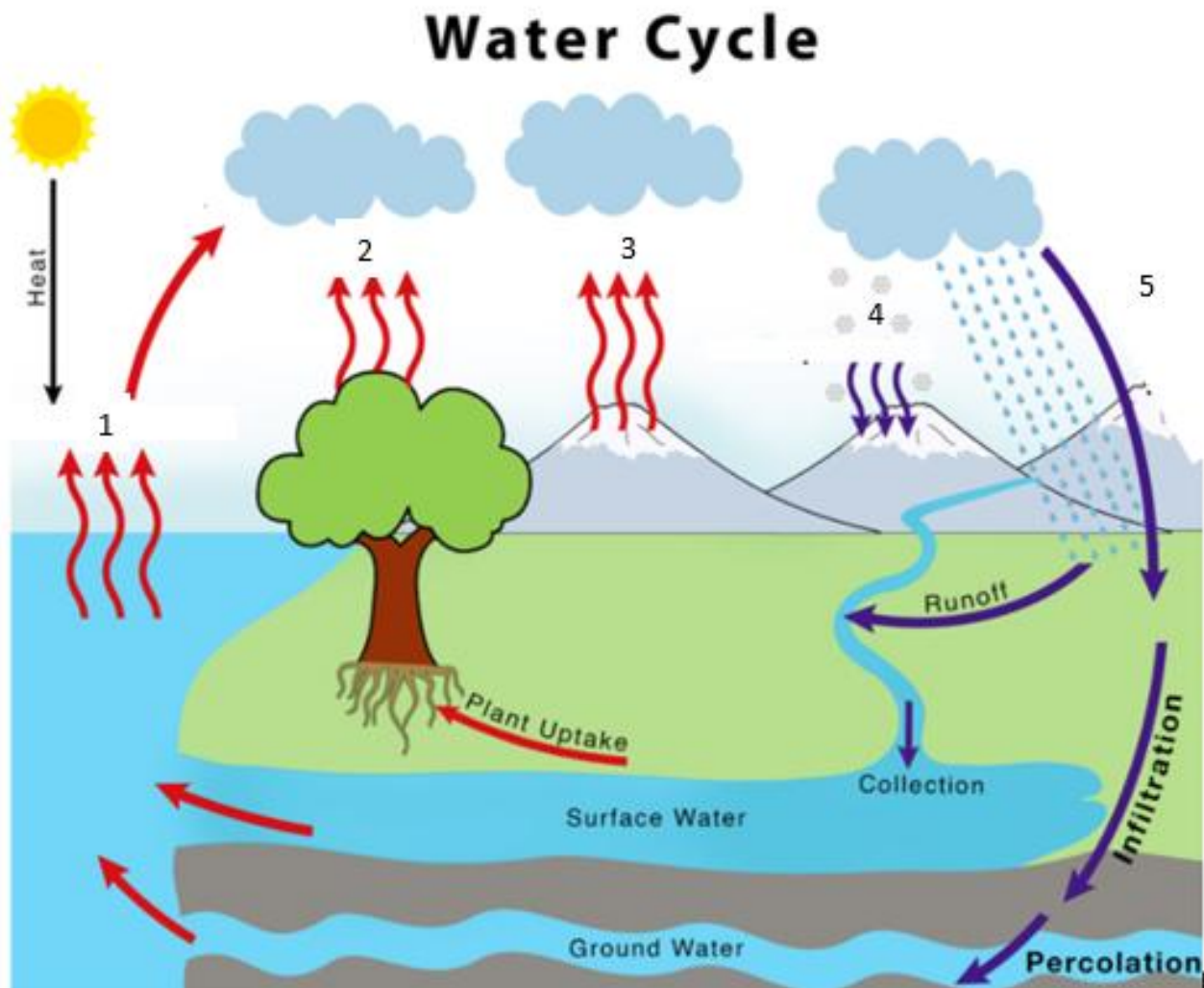
Test Score _____ (90) + Salinometer Testing Score _____ (10) = TOTAL SCORE _____ (100)

Part 1 – Freshwater Ecology

1. (1 point) What percent of the earth's water is freshwater?
A. 1% C. 10%
B. 3% D. 30%
2. (1 point) Some of the earth's freshwater is stored at the ice caps. How much of earth's water is usable by organisms as liquid water?
A. 1% C. 10%
B. 3% D. 30%
3. (1 point) A watershed is also called a
A. Water cycle C. Hectare
B. Point source D. Drainage basin
4. (6 points) Urbanization within a watershed (without mitigating Stormwater Best Management Practices) causes which of the following in downstream watercourses (circle one for each row):

A. An increase or decrease in the bankfull discharges?	Increase	Decrease
B. An increase or decrease in flood volumes?	Increase	Decrease
C. An increase or decrease in the time it takes for peak flow to occur?	Increase	Decrease
D. An increase or decrease in flow velocity?	Increase	Decrease
E. An increase or decrease in max particle size of eroded sediment?	Increase	Decrease
F. An increase or decrease in flow temperature?	Increase	Decrease
5. (1 point) The bankfull discharge, which defines the shape of a natural stream, is the peak flow from a rain event that occurs how frequently on average?
A. Once every 6 months C. Once every 5 years
B. Once every 1.5 years D. Once every 100 years
6. (1 point) Watershed management strategies include all of the following except:
A. Stormwater detention basins with outlets that control flood, bankfull, and/or first flush volumes
B. Stormwater Pollution Prevention Plans
C. Master Plans that incorporate land use and soil type data
D. National Flood Insurance Program
7. (1 point) Types of Point Source Pollution include all of the following EXCEPT:
A. Excess fertilizer from crops D. Oil spills
B. Effluent from power plants E. Stormwater drainage system outlet
C. Industrial discharges pipes
8. (1 point) Types of Non-Point Source Pollution include all of the following EXCEPT:
A. Acid rain D. Wastewater treatment plant discharges
B. Litter E. Atmospheric deposition of airborne
C. Septic systems contaminants

9. (1 point) When eroded river bank and bed material settles out of suspension, the resulting sedimentation harms aquatic life through all of the following mechanisms EXCEPT:
- A. Reduced photosynthesis
 - B. Impaired visibility
 - C. Increased dissolved oxygen levels
 - D. Increased water temperature
 - E. Pollutant transport
 - F. Eutrophication
10. (5 points) Identify the missing steps (#1-5) of the water cycle depicted below. (Hint: All five steps end in the suffix "tion".)



Step #1:

Step #4:

Step #2:

Step #5:

Step #3:

11. (1 point) The term "Nitrification" refers to
- A. The biological oxidation of ammonia to nitrate
 - B. Conversion of organic nitrogen to inorganic nitrogen by microorganisms
 - C. The process by which free nitrogen is converted into ammonia
 - D. the oxidation of nitrate into nitrite

12. (1 point) Which law states that growth of a plant is not controlled by the total amount of resources available, but by the scarcest resource (limiting factor) available?
- A. Sørensen's Law of Community
 - B. Liebig's Law of the Minimum
 - C. Jaccard's Law of Similarity
 - D. Kolmogorov's Law of 1941
13. (1 point) Of all the elements recycled in the biosphere, this element is the scarcest and is usually the limiting factor in any ecological system:
- A. Phosphorus
 - B. Nitrogen
 - C. Carbon
 - D. Calcium
14. (1 point) The Redfield-Richards ratio for freshwater systems describes
- A. the optimum ratio of carbon, nitrogen, and phosphate in freshwater
 - B. the optimum ratio of dissolved oxygen, nitrogen, and carbon dioxide in freshwater
 - C. the optimum ratio of plants, herbivore animals, and carnivore animals in a food chain
 - D. the optimum ratio of plants, herbivore animals, and carnivore animals in a food web
15. (1 point) The Redfield-Richards ratio for freshwater systems is
- A. 100:10:1
 - B. 106:16:1
 - C. 1:16:160
 - D. 1:6:116
16. (1 point) In a typical water treatment plant, the correct order of processes is
- A. Filtration, screening, coagulation and flocculation, disinfection, sedimentation, and distribution
 - B. Disinfection, screening, sedimentation, coagulation and flocculation, filtration, and distribution
 - C. Screening, coagulation and flocculation, sedimentation, filtration, disinfection, and distribution
 - D. Screening, sedimentation, filtration, disinfection, coagulation and flocculation, and distribution
17. (1 point) In the wastewater treatment process, what is Return Activated Sludge?
- A. The fraction of settled material that is reintroduced into the aeration tank to maintain the desired level of microorganisms to break down organic matter
 - B. The fraction of settled material that is collected from the bottom of the secondary clarifier and dried out before being landfilled
 - C. The fraction of settled material that is collected from the bottom of the aeration tank to be filtered through activated charcoal before being returned to the receiving waterbody
 - D. The fraction of settled material that is dried out and used as fertilizer for agricultural crops
18. (1 point) What is involved in the coagulation process?
- A. Filtering water through a sand or charcoal filter
 - B. Stirring water in a circular tank to create a vortex that forces suspended solids to the center
 - C. Adding chemical to dissolve suspended solids into the water
 - D. Adding chemicals to water to neutralize the charge of suspended solids

19. (1 point) What is involved in the flocculation process?
- A. Adding chemicals to water to increase the water's pH
 - B. Adding chemicals to water to decrease the water's pH
 - C. Adding chemicals to water to cause smaller particles to clump together
 - D. Adding chemicals to water to break up large particles into smaller ones
20. (1 point) The "floc" used in flocculation is
- A. Aluminum and magnesium
 - B. Calcium and magnesium
 - C. Lime and alum
 - D. Activated sludge
21. (1 point) All of the following are coagulants used in water treatment except:
- A. Aluminum sulfate
 - B. Sodium sulfate
 - C. Ferric chloride
 - D. Ferric sulfate
 - E. Sodium aluminate
 - F. Polyaluminum chloride
 - G. Ferrous sulfate

Part 2 – Freshwater Macroinvertebrates & Aquatic Nuisance Species

22. (6 points – Select all that apply) According to the EPA, benthic macroinvertebrates are reliable indicators of the biological condition of a freshwater waterbody because they
- A. spend all or most of their lives in water
 - B. are easy to collect
 - C. respond to human disturbance in fairly predictable ways
 - D. are highly tolerant of environmental pollutants
 - E. are relatively easy to identify in the laboratory
 - F. consume hydrocarbons
 - G. often live for more than a year
 - H. have limited mobility
23. (1 point) "Benthic" macroinvertebrates are those that
- A. are bottom dwellers
 - B. float on the water's surface
 - C. have wings before their adult stage
 - D. can both swim and fly
24. (1 point) From Most Tolerance to Least Tolerant, the pollution tolerance level of the four macroinvertebrates listed below is:
- A. Mayfly > Dragonfly > Blackfly > Horsefly
 - B. Dragonfly > Mayfly > Horsefly > Blackfly
 - C. Horsefly > Blackfly > Dragonfly > Mayfly
 - D. Blackfly > Dragonfly > Mayfly > Horsefly
25. (1 point) What do the E, P, and T stand for in "EPT Index"?
- A. Environmental Pollutant Total
 - B. Environmental Pathology Test
 - C. Ephemeroptera, Plecoptera, and Trichoptera
 - D. Epidemiological, Phytospheric, and Tropospheric

26. (1 point) A high EPT index for a waterbody indicates
- A. a high level of pollution within a waterbody
 - B. a high number of macroinvertebrates in the waterbody
 - C. a high density of pollution-tolerant macroinvertebrates in the waterbody
 - D. a high percentage of pollution-sensitive macroinvertebrates in the waterbody
27. (1 point) What do adult mayflies eat primarily?
- A. Green plant material that floats on top of the water's surface
 - B. Decaying woody material at the bottom of rivers and lakebeds
 - C. Almost anything (carbon-containing)
 - D. Nothing
28. (1 point) Damselfly larvae are aquatic predators in freshwater habitats that capture prey by shooting out their long, hinged lower lip called a
- A. ring
 - B. mask
 - C. tongue
 - D. whip
29. (4 points) All of the following are Class 1 – Pollution Sensitive freshwater adult stage macroinvertebrates EXCEPT (Circle the exception):



30. (1 point) According to the EPA, one of the most effective ways to control the local mosquito population is to:
- A. Burn citronella candles near lawns, within 15 feet of freshwater bodies
 - B. Use skin repellents containing DEET
 - C. Eliminate containers and debris that can hold rainwater from yards
 - D. Add sodium bicarbonate (baking soda) to stormwater detention basins to lower pH

31. (4 points) Circle the image below that shows a mayfly larva.



32. (1 point) Aquatic nuisance plants include all of the following EXCEPT:

- A. Pickerelweed
- B. Purple Loosestrife
- C. Eurasian Water Milfoil
- D. Water Hyacinth

33. (3 points) Circle the image below that shows Purple Loosestrife. (The other two images show Fireweed and Verbena Hastata, which are commonly mistaken for Purple Loosestrife.)



34. (1 point) Which of the following aquatic nuisance animals was first observed in the Canadian waters of Lake Erie in 1986?

- A. Asian Carp
- B. Asian Tiger Mosquito
- C. Spiny Water Flea
- D. Zebra Mussel

35. (3 points) Circle the image that shows a zebra mussel. (The other two images show a quagga mussel and a unionid clam, both native Michigan species.)



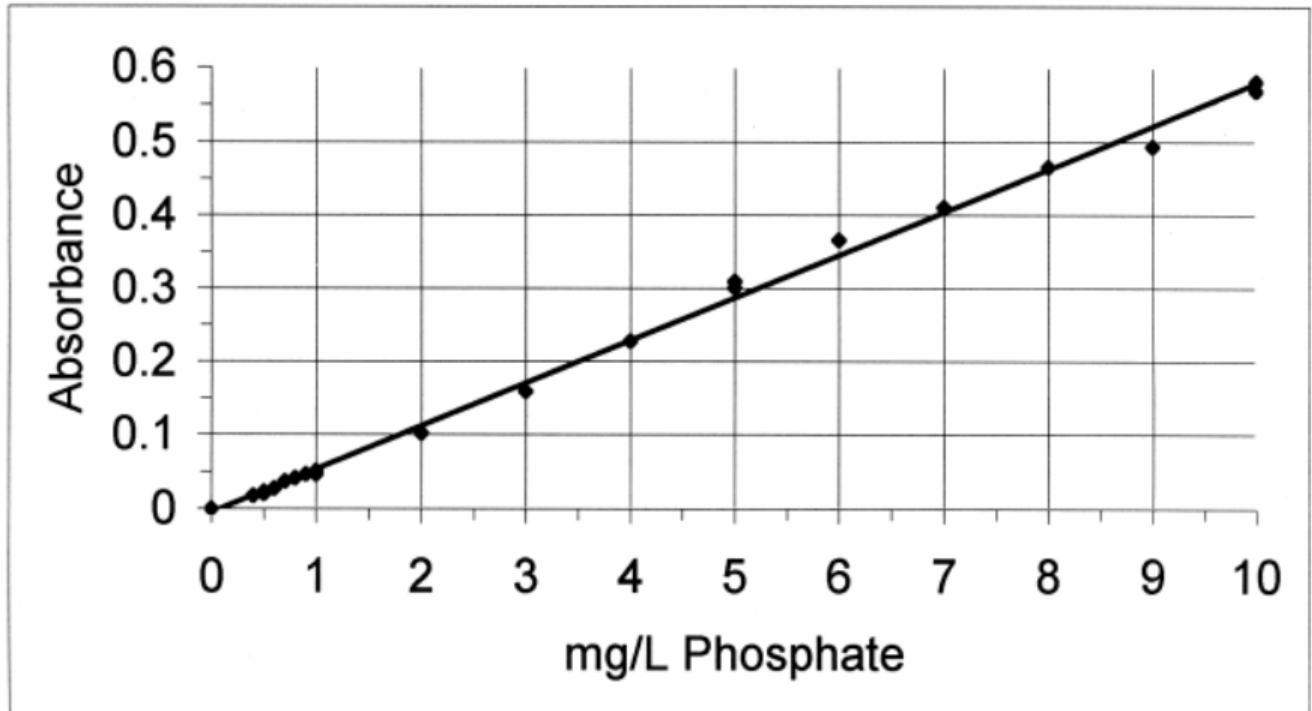
36. (1 point) According to the International Institute for Sustainable Development, one effect of the increasing zebra mussel population within the Great Lakes has been that
- A. the phytoplankton and zooplankton populations have increased
 - B. the Secchi depth has decreased
 - C. the water in the upper levels of the lakes has gotten colder
 - D. Lake Trout have migrated to deeper waters

Part 3 – Water Monitoring and Analysis

37. (1 point) The salinity of Lake Michigan is between
- A. 10 to 50 parts per billion
 - B. 50 to 100 parts per billion
 - C. 10 to 50 parts per million
 - D. 50 to 500 parts per million

38. (1 point) One part per million is equivalent to
- A. 0.01 percent
 - B. 0.001 percent
 - C. 0.0001 percent
 - D. 0.00001 percent
39. (1 point) Dissolved oxygen
- A. Increases with increasing water temperature
 - B. Decreases with increasing water temperature
 - C. Is independent of water temperature
 - D. Causes water temperature to increase
40. (1 point) The two stages of BOD are
- A. Aerobic and anaerobic demand
 - B. Chemical and biological demand
 - C. Carbonaceous and nitrification demand
 - D. Natural and artificial demand
41. (1 point) Assuming $k_{20} = 0.1$ per day, if the 3-day BOD of a wastewater sample at 20 degrees C is 400 mg/l, then the ultimate BOD is
- A. 398 mg/l
 - B. 548 mg/l
 - C. 802 mg/l
 - D. 1202 mg/l
42. (1 point) Assuming $k_{20} = 0.1$ per day, if the 3-day BOD of a wastewater sample at 20 degrees C is 400 mg/l, then the BOD_5 (at 20 degrees C) is
- A. 398 mg/l
 - B. 548 mg/l
 - C. 802 mg/l
 - D. 1202 mg/l
43. (1 point) Turbidity is a measure of
- A. how acidic the water is
 - B. how cloudy the water is
 - C. how much dissolved oxygen the water holds
 - D. the water's capacity to resist a change in pH
44. (2 points – Select all that apply) Turbidity can be measured using a
- A. Salinometer
 - B. Spectrophotometer
 - C. Cone Penetrometer
 - D. Secchi disk
45. (2 points – Select all that apply) Turbidity is measured in
- A. Milligrams per liter (mg/l)
 - B. Parts per million (ppm)
 - C. Formazin Nephelometric Units (FNU)
 - D. Nephelometric Turbidity Units (NTU)
46. (1 point) Phosphorus concentrations reported as PO_4 can be converted to concentrations of P by
- A. Multiplying by five
 - B. Dividing by five
 - C. Multiplying by three
 - D. Dividing by three

47. (1 point) The ideal (total) phosphorus concentration for freshwater systems is generally
- A. Below 0.5 ppm
 - B. Above 5 ppm
 - C. Below 50 ppm
 - D. Between 0.5 and 5 ppm
48. (1 point) A standard curve showing the concentration of absorbance and phosphate concentration (below) was developed. Your water sample has an absorbance of 0.2 AU. What is the estimated Phosphate concentration of your sample?
- A. 3.5 mg/L Phosphate
 - B. 3 ppm Total phosphorus
 - C. 350 ppb Phosphate
 - D. 3000 ppb Total phosphorus



49. (2 points – Select all that apply) Dissolved organic carbon (DOC) is the fraction of organic carbon that:
- A. can pass through a filter with a pore size between 0.22 and 0.7 micrometers
 - B. can settle out of suspension within 24 hours
 - C. does not contribute to turbidity due to its extremely small size
 - D. attenuates light which can alter vertical temperature profiles
50. (1 point) A spectrophotometer is a device that can be used to measure
- A. acidity
 - B. turbidity
 - C. salinity
 - D. dissolved oxygen
51. (1 point) Although generally not pathogenic, the presence of which of the following indicates a pathway for human pathogens (ex. water borne Viruses, bacteria, protozoa) to enter the water source?
- A. Nitrates
 - B. Phosphates
 - C. Salinity
 - D. Fecal coliform

52. (1 point) How much oxygen can water hold at 70 degrees Fahrenheit?
A. 8.9 mg/l B. 14.6 mg/l C. 21 mg/l D. 24.6 mg/l
53. (1 point) Does dissolved oxygen (all other factors being equal) increase or decrease with increasing depths?
A. Increase B. Decrease
54. (1 point) Does dissolved oxygen (all other factors being equal) increase or decrease with increasing salinity?
A. Increase B. Decrease
55. (1 point) Does dissolved oxygen (all other factors being equal) increase or decrease with increasing altitude?
A. Increase B. Decrease
56. (1 point) Is dissolved oxygen (all other factors being equal) higher during the daytime or the nighttime?
A. Daytime B. Nighttime
57. (1 point) Different species have varying levels of tolerance to low dissolved oxygen levels. Trout and carp have the following minimum requirements for dissolved oxygen levels, respectively:
A. 5 mg/l; 2 mg/l C. 1 mg/l; 6 mg/l
B. 3 mg/l; 5 mg/l D. 6 mg/l; 0.5 mg/l
58. (2 points – Select all that apply) The nutrients primarily responsible for eutrophication are
A. Nitrogen D. Phosphorus
B. Methane E. Hydrogen
C. Sodium F. Fluoride
59. (1 point) Which of the following develops on slopes, in depressions, or on flats as a result of sustained flows of mineral-rich groundwater in the root zone, which creates persistent anaerobic conditions that limit the decomposition of plant debris?
A. Fen C. Cartesian well
B. Bog D. Aquifer
60. (1 point) If you took water samples from both a bog and a fen, which one would you expect to have the higher calcium concentration?
A. Fen B. Bog
61. (1 point) Which of the following is often measured in “percent saturation”?
A. Salinity C. Fecal coliform
B. Dissolved oxygen D. Turbidity
62. (1 point) The ideal pH range for freshwater aquatic life is
A. 3-5 B. 5-6.5 C. 6.5-8.5 D. 8-10